

AstroTracking AI

Real-Time Asteroid Hazard Prediction & Astronomy Research Platform

Complete Project Report

1. Project Overview

AstroTracking AI is a full-stack web application that combines real-time NASA Near-Earth Object (NEO) data, machine-learning-style risk classification, exoplanet light-curve simulation, and an AI-powered astronomy chatbot. It serves researchers, students, and space enthusiasts with an interactive cosmic dashboard.

2. Core Modules

Mission Control Dashboard — Live asteroid feed from NASA NEO API with filtering, pagination (12/page), risk meters, and detailed modals.

Risk Classification Engine — Deterministic hazard scoring algorithm classifying asteroids by size, velocity, miss-distance, and orbit class.

Light Curve Simulation — Exoplanet transit modeling with stellar flux generation based on Kepler-style parameters.

Research Dashboard — Kepler simulation metrics, dataset visualization, and educational About sections explaining scientific workflows.

AstroBot AI Chatbot — Gemini-2.5-flash-lite powered conversational AI with NASA Image API integration, voice dictation (Web Speech API), and AI autocorrect.

Contact Mission Control — Deep-space themed contact form persisting messages directly to the backend.

Developer Showcase — Editable team profile section with social links.

3. Technology Stack

Frontend: React 18, TypeScript 5, Vite 5, Tailwind CSS v3, shadcn/ui, Framer Motion, React Router, TanStack Query, Recharts.

Backend: Lovable Cloud (Supabase) — PostgreSQL with RLS, Edge Functions (Deno), Auth, Storage.

AI: Lovable AI Gateway with google/gemini-2.5-flash-lite for chat & voice autocorrect.

External APIs: NASA NEO Feed, NASA Image and Video Library — proxied via Edge Function with DEMO_KEY fallback.

4. Design System

Cosmic dark-space aesthetic: HTML5 Canvas StarField background, glassmorphism cards, Orbitron display font, semantic HSL design tokens, custom cursor star-trail effect, and responsive layouts tuned for mobile, tablet, and desktop.

5. Key Features

- Real-time NASA NEO data with auto-refresh
- Deterministic ML-style risk scoring (Critical / High / Medium / Low)
- Interactive stat cards triggering drill-down modals
- AI chatbot with voice input + auto-correct of speech transcripts
- Visual microphone state (red pulsing while listening)
- Custom star-trail cursor across the entire app
- Live date widget in headers
- State-based root navigation (no global Home button)
- 12-items-per-page pagination with filter-reset rules
- Educational About sections on every research page
- Editable Developer profiles backed by the database
- Full responsive design + secure HTTPS-only voice features

6. Architecture

Routing: React Router with state-based dashboard rendering on the root path. Sub-pages use a Back button rather than a global Home button.

Edge Functions:

- *nasa-proxy* — secures the NASA API key and provides DEMO_KEY fallback.
- *astronomy-chat* — handles chat completions, NASA image search, and voice transcript autocorrect via Gemini.

Database: Tables for developer_info and contact messages with strict Row-Level Security policies.

Security: All API keys stored as edge secrets; no client-side credentials; HTTPS enforced for microphone access.

7. Performance & Scalability

Built on Lovable Cloud's auto-scaling Supabase infrastructure and globally distributed edge runtime. Vite produces optimized bundles with code-splitting; TanStack Query caches NASA responses to reduce upstream load. The proxy layer prevents API-key leakage and rate-limit exhaustion.

8. Voice Interaction

Web Speech API powers single-utterance dictation with secure-context checks, locale-aware recognition (navigator.language), graceful error handling for no-speech / network / hardware issues, and AI-driven autocorrect that fixes technical astronomy terms (e.g., 'black whole' → 'black hole', 'nassa' → 'NASA').

9. Pages

/ — Landing hero + Mission Control dashboard

/chatbot — AstroBot conversational AI

/light-curve — Exoplanet transit simulation

/research — Kepler research dashboard

/contact — Contact Mission Control form

/developed-by — Team / developer showcase

10. Future Enhancements

- User authentication for personalized watchlists
- WebSocket real-time alerts for newly-discovered NEOs
- 3D orbital visualization with Three.js
- Historical asteroid trajectory playback
- Mobile PWA with offline NEO cache

